

SAKON NAKHON, Thailand

WMO: 483560

Lat: 17.160N

Lon: 104.133E

Elev: 564

StdP: 14.40

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 55.4	(c) 58.1	(d) 45.9	(e) 46.6	(f) 64.0	(g) 48.7	(h) 52.0	(i) 65.0	(j) 11.8	(k) 76.3	(l) 10.5	(m) 75.8	(n) 0.7	(o) 120	(p) 0.295

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	16.8	99.3	77.0	96.9	77.2	94.7	77.6	81.6	90.0	80.9	89.3	80.4	88.6	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	157.3	85.0	79.0	153.8	84.4	78.4	150.7	83.9	45.9	90.2	45.2	89.6	44.6	88.9	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.9	(c) 7.6	(e) 50.1	(f) 102.5	(g) 3.1	(h) 2.2	(i) 47.9	(j) 104.1	(k) 46.1	(l) 105.4	(m) 44.3	(n) 106.6	(o) 42.0	(p) 108.2		
(4) 10.0	(5) 8.9	(6) 7.6	(7) 50.1	(8) 102.5	(9) 3.1	(10) 2.2	(11) 47.9	(12) 104.1	(13) 46.1	(14) 105.4	(15) 44.3	(16) 106.6	(17) 42.0	(18) 108.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	73.3	76.9	82.1	85.3	84.6	84.0	82.8	82.3	82.1	80.9	77.7	73.2
	DBStd	5.69	4.87	5.91	5.80	4.68	3.41	2.68	2.77	2.34	2.22	2.65	3.96	4.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	10	4	2	0	0	0	0	0	0	0	0	0	4
	CDD50	11111	721	755	995	1059	1072	1020	1018	1001	962	957	831	720
	CDD65	5647	260	337	530	609	607	570	553	536	512	492	381	260
	CDH74	59816	2143	3243	6063	7555	7113	6590	5952	5554	5223	4853	3410	2117
	CDH80	24394	712	1427	3041	3978	3205	2676	2105	1795	1736	1795	1263	661
(14) Wind	WSAvg	1.8	2.2	2.4	2.4	1.9	1.5	1.5	1.7	1.7	1.2	1.6	1.8	2.0
(15) Precipitation	PrecAvg	63.6	0.2	0.7	1.9	3.1	9.7	11.8	8.6	13.7	10.4	2.9	0.4	0.2
	PrecMax	79.3	1.9	3.7	5.5	5.0	17.9	17.7	14.3	39.3	19.7	10.6	2.3	1.8
	PrecMin	53.7	0.0	0.0	0.0	0.6	5.4	5.8	2.2	4.2	4.3	0.1	0.0	0.0
	PrecStd	8.3	0.5	1.1	1.6	1.3	3.4	3.7	4.0	8.2	4.6	2.7	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	97.6	101.7	103.3	99.7	96.1	95.2	93.2	92.4	92.0	93.5	91.3
		MCWB	72.0	72.6	73.8	77.2	79.3	79.9	80.1	80.4	79.4	77.3	76.1	74.6
	2%	DB	89.1	94.3	98.6	100.2	96.5	93.7	92.1	91.2	90.7	90.5	90.7	88.6
		MCWB	71.6	73.1	75.1	77.7	79.6	79.9	79.6	79.7	78.9	76.8	75.2	72.9
	5%	DB	86.7	91.5	95.9	97.3	94.1	91.9	90.2	89.4	89.3	89.3	88.8	86.3
		MCWB	70.9	73.1	75.5	78.1	79.4	79.6	79.4	79.4	78.4	76.4	74.4	71.5
	10%	DB	84.1	88.6	93.0	94.6	91.7	90.1	88.3	87.7	87.8	88.0	86.8	83.9
		MCWB	69.7	72.6	75.5	77.9	79.0	79.4	79.1	79.1	78.0	75.8	73.4	70.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	78.2	79.8	81.3	82.0	82.3	81.9	81.9	81.7	80.0	78.6	76.6
		MCDB	86.3	89.3	92.3	94.5	92.4	90.4	89.6	89.0	87.2	87.8	87.6	87.4
	2%	WB	74.4	76.5	78.3	80.2	81.1	81.3	80.9	80.9	80.6	79.0	77.4	74.6
		MCDB	84.0	88.4	91.2	93.6	91.2	89.3	88.2	87.7	87.1	86.4	86.2	85.1
	5%	WB	72.6	75.1	77.4	79.5	80.4	80.6	80.1	80.1	79.9	78.1	76.4	73.1
		MCDB	82.6	86.4	90.4	92.5	90.2	88.7	87.1	86.6	86.0	85.3	84.8	83.6
	10%	WB	71.0	73.8	76.4	78.7	79.8	80.0	79.6	79.6	79.2	77.4	75.2	71.6
		MCDB	82.0	84.7	89.0	91.2	88.8	87.5	86.2	85.7	85.0	84.3	83.1	81.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	21.2	19.0	16.8	13.4	11.5	10.3	10.3	11.0	14.0	17.7	20.2
		MCDBR	22.8	23.5	21.2	19.4	16.3	13.8	12.9	12.8	13.2	15.1	18.8	20.8
		MCWBR	8.0	7.5	6.3	5.5	4.9	4.3	4.2	4.6	4.1	4.6	5.9	7.1
	5% WB	MCDBR	19.7	19.7	18.6	17.3	14.2	12.5	11.6	11.8	11.7	13.3	15.7	19.3
		MCWBR	8.0	7.4	6.6	5.9	5.2	4.7	4.6	4.8	4.5	4.5	5.6	7.2
(40) Clear-Sky Solar Irradiance	taub		0.457	0.569	0.662	0.628	0.511	0.470	0.457	0.468	0.506	0.519	0.456	0.454
	taud		2.146	1.855	1.683	1.799	2.146	2.289	2.321	2.278	2.159	2.087	2.237	2.178
	Ebn at Noon		261	237	220	229	254	262	266	265	255	248	260	258
	Edh at Noon		46	65	79	71	49	43	41	43	49	51	42	44
(44) All-Sky Solar Radiation	RadAvg		1538	1629	1628	1747	1674	1548	1437	1430	1471	1584	1551	1474
(45) RadStd			91	105	69	87	112	118	137	104	115	112	57	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page